Math 420 - Exercise Set 2 (9 .10.2026)

1. Let $(X, \mathcal{T})$ be a topological space and $\mathcal{B}$ be a base for $\mathcal{T}$. Show that $\mathcal{B}$ is a subbase for $\mathcal{T}$.

2. Let X be a set and $x_0,\ y_0 \in X$ be distinct fixed points. If

$$\mathcal{T}_1 = \{U \mid U \subseteq X \text{ and } x_0 \in U\} \cup \{\emptyset\}$$

and

$$\mathcal{T}_1 = \{V \mid V \subseteq X \text{ and } y_0 \in V\} \cup \{\emptyset\}$$

then show that

(i) $(X, \mathcal{T}_1)$ and $(X, \mathcal{T}_2)$ are homeomorphic.

(ii) $\mathcal{T}_1$ and $\mathcal{T}_2$ are not comparable.

3. Let $X = \{a, b, c, d, e\}$ and let $\mathcal{S} = \{\{a, c\}, \{b\}, \{c, d, e\}\}$ be a class of subsets of X.

(i) Determine the topology $\mathcal{T}$ on X generated by $\mathcal{S}$.

(ii) Determine if the following sequences are convergent in $(X, \mathcal{T})$.

(a) $(x_n) = (a, c, a, c, a, c, \cdots)$,

(b) $(y_n) = (c, d, c, d, c, d, \cdots)$,

(c) $(z_n) = (a, c, d, a, c, d, a, c, d, \cdots)$.

(iii) Determine if f and g defined by the following diagrams are continuous :



